Atharva Mendhulkar

+91 8591436357 | mendhu36@outlook.com | linkedin.com/in/mendhu36 | github.com/Atharva-Mendhulkar | portfolio

Education

Vellore Institute of Technology, Vellore Jul. 2024 – Aug. 2028
B.Tech in Information Technology

Experience

Avesta Labs Ahmedabad, India
AI Engineer Intern *May 2026 – Present*
- Built realtime AI voice agents using LiveKit, OpenAI APIs, Temporal, Redis, and Kubernetes for production conversational workflows.
- Implemented distributed orchestration pipelines, REST APIs, WebSockets, JWT authentication, and observability systems for concurrent AI workloads.
- Achieved **530ms median** and **sub-700ms P95** conversational latency in production deployments.

Indian Institute of Technology Gandhinagar & Maker Bhavan Foundation Gandhinagar, India
Summer Intern (InventX Scholar) *May 2025 – Jul. 2025*
- Selected in the **top 1%** of 2000+ applicants for a competitive research and embedded systems innovation program.
- Designed algorithmic control logic for a patent-pending assistive technology system integrating hardware sensors with computational pipelines.

Research

Physics-Informed Neural Networks for Urban Air Quality Modeling Dec. 2025 – Present
- Designed a scientific machine learning framework enforcing atmospheric advection–diffusion PDE constraints for large-scale spatio-temporal air quality prediction.
- Integrated sparse environmental sensor observations with ERA5 meteorological fields using probabilistic modeling techniques for physically consistent inference.
- Achieved $\mathbf{R^2} \approx \mathbf{0.85}$ on temporal holdout evaluation while outperforming standard deep learning baselines under sparse-data conditions.
- Applied Monte Carlo Dropout uncertainty quantification to estimate predictive confidence in low-observation regions.

Dynamic Graph Colouring using Temporal Graph Neural Networks | Paper Sep. 2025 – Nov. 2025
- Designed a spatio-temporal graph reasoning architecture for online graph colouring under dynamic edge updates and evolving network topology.
- Developed a budget-constrained repair algorithm minimizing recolouring operations while preserving computational feasibility guarantees.
- Optimized a dual-objective loss balancing conflict minimization and temporal stability constraints.
- Achieved **35% fewer recolours** and **12% lower inference latency** compared to baseline approaches.

Projects

AVARA — Autonomous Validation & Agent Risk Authority | GitHub 2026
- Built a runtime governance and validation control plane for autonomous AI agents operating in production environments.
- Designed a zero-trust sidecar gateway intercepting agent execution requests and enforcing permission scopes, execution policies, and identity validation.
- Implemented runtime anomaly detection and execution circuit breakers for adversarial tool use, recursive execution loops, and unsafe autonomous actions.
- Maintained **sub-15ms** decision latency under concurrent agent workloads with negligible execution overhead.

Floework — Productivity & Collaboration Platform | GitHub | Demo 2026
- Built a multi-tenant SaaS collaboration platform enabling teams to manage projects, tasks, and realtime communication across **50+ isolated workspaces** with support for **10,000+ task mutations**.
- Engineered a PostgreSQL, Redis, and BullMQ backend with **38 APIs**, processing **25,000+ asynchronous jobs**.
- Implemented WebSocket-based realtime synchronization with optimistic concurrency control, sustaining **1,200+ concurrent users** at **45ms average** and **85ms P95** synchronization latency.

AEOS — Autonomous Enterprise Operations System | GitHub 2026
- Built an autonomous enterprise operations platform that coordinates AI agents to manage incident response, operational workflows, compliance validation, and failure recovery from logs, documents, screenshots, PDFs, audio, and other enterprise data sources.
- Architected a distributed system comprising **13 microservices**, **18 Dockerized services**, and **54 API endpoints**, enabling **9 concurrent specialized agents** to collaboratively execute long-running enterprise workflows.
- Implemented asynchronous orchestration, runtime governance, observability, audit logging, and Redis Pub/Sub coordination using FastAPI, PostgreSQL, Docker, and WebSockets, supporting **7 multimodal input types** and validated across **13 enterprise workflow simulations**.

K-PHD — Kernel-level Predictive Hang Detector | GitHub 2026
- Built a Linux kernel module for predictive process hang detection using scheduler instrumentation, O(1) kernel hash tables, and Generic Netlink IPC for real-time kernel-to-userspace communication.
- Processed **5M+ scheduler events** while tracking **5,000+ concurrent processes**, maintaining **<0.5% kernel overhead** and sub-millisecond telemetry export latency.
- Predicted application hangs up to **2.8s before system watchdog timeouts**, achieving **94% precision** with a **4% false-positive rate** across **100+ stress-test scenarios**.

Patent & Innovation

UPRS — Universal Process Responsiveness System | Patent Application No. **202641054886**

- Engineered a predictive operating-system-level hang detection framework using low-level kernel telemetry and lightweight inference models.
- Predicted application hangs **0.5–5s** before occurrence, achieving **92.5% precision** and **85.6% recall**.
- Reduced user-visible UI freezes by **82%** and validated cross-platform deployment across Linux and macOS.

Technical Skills

Languages: Python, C/C++, Bash, SQL, JavaScript, TypeScript
Systems & Infrastructure: Linux/Unix Internals, Operating Systems, Distributed Systems, Docker, Redis, PostgreSQL, Kubernetes
AI & Reasoning Systems: LLM Agents, Autonomous AI Systems, Runtime Validation, Evaluation Pipelines, Prompt Engineering
Core CS: Data Structures & Algorithms, System Design, DBMS, Computer Networks, Object-Oriented Programming

Achievements

Samsung PRISM Metaverse 2.0 Finalist — Top 20 out of 450+ participants.

Goldman Sachs India Hackathon '26 — **Rank 26** among 15,000+ participants.

CodeChefVIT DevSoc'26 Winner — Digital Economy Track (500+ participants).

SIH 2025 Semi-finalist — Top teams among 400+ participants.

Innovation Week 2025 — Presented an AI and IoT-based maternal healthcare platform to scientists and researchers from ISRO and DRDO.